Sets

1.What is set?

A set is a well-defined collection of distinct objects.

Examples:

- $A = \{1, 2, 3, 4, 5\}$
- $B = \{a, e, i, o, u\}$
- $C = \{2, 4, 6, 8\}$

2. Types of Sets

Type	Meaning	Example
Empty set	No elements	$\emptyset$
Singleton set	One element	$\{5\}$
Finite set	Limited elements	$\{1, 2, 3\}$
Infinite set	Unlimited elements	$\{1, 2, 3, \ldots\}$
Equal sets	Same elements	$\{1, 2\}, \{2, 1\}$
Subset	Every element belongs to another set	$\{1, 2\} \subseteq \{1, 2, 3\}$

3. Important Set Symbols

Symbol	Meaning
$\in$	Belongs to
$\notin$	Does not belong to
$\subseteq$	Subset
$\cup$	Union
$\cap$	Intersection
$\emptyset$	Empty set
A' or A^c	Complement
$A - B$	Difference
U	Universal set

4. Set Operations

Take these two sets:

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

Union – $A \cup B$

All unique elements from both sets.

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

Intersection – $A \cap B$

Common elements only.

$$A \cap B = \{3, 4\}$$

Difference – $A - B$

Elements present in A but not in B.

$$A - B = \{1, 2\}$$

Difference – $B - A$

$$B - A = \{5, 6\}$$

5. Cardinality of a Set

Cardinality means number of elements in a set.

$$|A| = 4, \quad |B| = 4$$

Very important GATE formula:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example:

- $|A| = 20$
- $|B| = 15$
- $|A \cap B| = 5$

Then:

$$|A \cup B| = 20 + 15 - 5 = \boxed{30}$$

6. Power Set

Power set means the set of all possible subsets.

If:

$$A = \{1, 2\}$$

Then:

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Number of subsets formula:

$$\boxed{|P(A)| = 2^n}$$

Here n = number of elements.

For 3 elements, total subsets = $2^3 = 8$.

Mini pre-study test

5-question diagnostic

Q1. If A = {1,2,3} and B = {3,4,5}, what is A ∪ B?

- A. {3}
- B. {1,2,3,4,5}
- C. {1,2}
- D. {4,5}

Q2. If A = {1,2,3} and B = {2,3,4}, what is A ∩ B?

- A. {1,4}
- B. {1,2,3,4}
- C. {2,3}
- D. ∅

Q3. How many subsets does a set with 4 elements have?

- A. 4
- B. 8
- C. 16
- D. 32

Q4. If |A| = 10, |B| = 8 and |A ∩ B| = 3, find |A ∪ B|.

- A. 15
- B. 18
- C. 21
- D. 24

Q5. Which statement is always true?

- A. A ∪ ∅ = ∅
- B. A ∩ ∅ = A
- C. A ∪ ∅ = A
- D. A − A = A

Next: Part 6 – Set Operations

Let's take:

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

1. Union ($A \cup B$)

All distinct elements from both sets.

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

2. Intersection ($A \cap B$)

Only common elements.

$$A \cap B = \{3, 4\}$$

3. Set Difference ($A - B$)

Elements in A but not in B.

$$A - B = \{1, 2\}$$

Similarly,

$$B - A = \{5, 6\}$$

Important: $A - B$ and $B - A$ are generally different.

4. Complement (A^c)

Suppose:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4\}$$

Then:

$$A^c = U - A = \{5, 6, 7, 8\}$$

Round 2 – Set operations

Q1. A = {1,2,3}, B = {3,4,5}. Find A ∪ B.

- A. {3}
- B. {1,2,3,4,5}
- C. {1,2}
- D. {4,5}

Q2. For the same A and B, find A ∩ B.

- A. {1,2}
- B. {3}
- C. {4,5}
- D. ∅

Q3. Find A − B.

- A. {1,2}
- B. {3}
- C. {4,5}
- D. {1,2,3,4,5}

Q4. If U = {1,2,3,4,5,6} and A = {2,4,6}, find A^c.

- A. {2,4,6}
- B. {1,3,5}
- C. {1,2,3}
- D. ∅

Part 7 – Cardinality of Sets

Cardinality means the number of distinct elements in a set.

Example:

$$A = \{2, 4, 6, 8, 10\}$$

$$|A| = 5$$

Here, $|A|$ means the size of set A.

1. Inclusion-Exclusion Principle

Very important for GATE

For two sets:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Why do we subtract the intersection?

Suppose:

- 20 students know Python.
- 15 students know SQL.
- 5 students know both.

If we add 20 + 15, the 5 students who know both are counted twice.

So subtract 5 once:

$$|A \cup B| = 20 + 15 - 5$$

$$\boxed{30}$$

2. Other useful formulas

For two sets:

$$|A - B| = |A| - |A \cap B|$$

$$|B - A| = |B| - |A \cap B|$$

For complement:

$$|A^c| = |U| - |A|$$

For disjoint sets:

$$A \cap B = \emptyset$$

Therefore:

$$|A \cup B| = |A| + |B|$$

3. GATE-style example

In a class of 100 students:

- 60 know Python.
- 45 know SQL.
- 25 know both.

Find:

1. Students who know at least one.
2. Students who know only Python.
3. Students who know only SQL.
4. Students who know neither.

Step 1: At least one

$$|P \cup S| = 60 + 45 - 25 = 80$$

Step 2: Only Python

$$60 - 25 = 35$$

Step 3: Only SQL

$$45 - 25 = 20$$

Step 4: Neither

$$100 - 80 = 20$$

Final answer

Category	Students
Only Python	35
Only SQL	20
Both	25
Neither	20
At least one	80

Check: Total = 100

Cardinality – 5 questions

Q1. If $|A|=30$, $|B|=20$ and $|A \cap B|=10$, find $|A \cup B|$.

- A. 40
- B. 50
- C. 60
- D. 30

Q2. In a group of 80 students, 50 know Python, 40 know SQL and 20 know both. How many know neither?

- A. 10
- B. 20
- C. 30
- D. 40

Q3. If $|U|=100$ and $|A|=65$, find $|A^c|$.

- A. 35
- B. 65
- C. 100
- D. 25

Q4. If $|A|=25$ and $|A \cap B|=8$, find $|A-B|$.

- A. 17
- B. 25
- C. 33
- D. 8

Q5. If A and B are disjoint, |A|=12 and |B|=18, find |A∪B|.

- A. 6
- B. 30
- C. 216
- D. 18

Part 9 – Set Identities

Set identities are rules that help simplify mathematical expressions involving sets.

Assume A, B, C are sets.

1. Identity laws

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Meaning: Union with an empty set or intersection with the universal set doesn't change A.

2. Domination laws

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

3. Idempotent laws

$$A \cup A = A$$

$$A \cap A = A$$

4. Complement laws

$$A \cup A^c = U$$

$$A \cap A^c = \emptyset$$

5. Commutative laws

Order doesn't matter.

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

6. Associative laws

Grouping doesn't matter.

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

7. Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Part 10 – De Morgan's Laws

These two laws are especially important.

First law:

$$\boxed{(A \cup B)^c = A^c \cap B^c}$$

Second law:

$$\boxed{(A \cap B)^c = A^c \cup B^c}$$

Worked example

Let:

$$U = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 3\}$$

$$B = \{3, 4\}$$

Find $(A \cup B)^c$.

First:

$$A \cup B = \{1, 2, 3, 4\}$$

Then:

$$(A \cup B)^c = \{5, 6\}$$

Using De Morgan's law:

$$A^c = \{4, 5, 6\}$$

$$B^c = \{1, 2, 5, 6\}$$

$$A^c \cap B^c = \{5, 6\}$$

Both methods give the same answer!

Q1. Simplify $A \cup (A \cap B)$.

- A. A
- B. B
- C. $A \cap B$
- D. $A \cup B$

Q2. Which is equal to $(A \cup B)^c$?

- A. $A^c \cup B^c$
- B. $A^c \cap B^c$
- C. $A \cap B$
- D. $A \cup B$

Q3. Simplify $A \cap (A \cup B)$.

- A. A
- B. B
- C. $A \cup B$
- D. $\varnothing$

Q4. Which identity is always true?

- A. $A \cup A^c = \varnothing$
- B. $A \cap A^c = U$
- C. $A \cup A^c = U$
- D. $A \cap U = \varnothing$

Q5. $(A \cap B)^c$ is equal to:

- A. $A^c \cap B^c$
- B. $A^c \cup B^c$
- C. $A \cap B$
- D. $A \cup B$

Full topic assessment

0/6 completed

MCQ: select one · MSQ: select all that apply · NAT: enter a number

Question 1

MCQ

A set A has 6 elements. How many proper subsets does it have?

- A. 32
- B. 63
- C. 64
- D. 62

Question 2

MCQ

If $|A|=40$, $|B|=30$ and $|A \cup B|=55$, find $|A \cap B|$.

- A. 10
- B. 15
- C. 20
- D. 25

Question 3 MSQ

Which of the following are always true?

- A. $A \cup \varnothing = A$
- B. $A \cap U = A$
- C. $A \cup A^c = U$
- D. $A \cap A^c = A$

Question 4 MSQ

Which statements about power sets are true?

- A. Every set is a subset of itself
- B. Empty set belongs to every power set
- C. A set with n elements has 2^n subsets
- D. $P(\varnothing) = \varnothing$

Question 5 NAT

A set has 7 elements. How many non-empty proper subsets does it have?

Enter numerical answer

Question 6 NAT

In a group of 120 students, 70 know Python, 55 know SQL and 25 know both. How many know neither?

Enter numerical answer